

A First Look at Ethics in Computer and Data Sciences

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2021-07-21

Student Materials

- worksheet
- Markdown

Overview

Summary: In a previous lab, ethics was defined as “the study of what it means to ‘do the right thing’”. The question then, is how do we know what is the right thing? This lab introduces the 3 ethical frameworks used in future labs. It begins to consider the virtue ethics framework at a very high level along with introducing the use of the other frameworks.

Ethics background required: None

Subject matter referred to in this lab: computer and data science tasks

Placement in overall ethics curriculum:

- If the first year groundwork sequence is incorporated into a program, this lab should be assigned as homework after the “Why Ethics?” lab and prior to completing the Virtue Ethics lab in class.
- If other labs are being used as stand-alone labs, this is a good homework assignment to use to introduce students to the ethics frameworks prior to completing any other lab.

Time required:

- As homework: 25 minutes

Learning objectives:

- Students are introduced to the 3 ethical frameworks that are considered in future labs
- Students become familiar with virtue-terms
- Students understand that virtues are characteristics that can be the basis for wise decisions

- Students try to apply one or more frameworks to make a decision about an ethical dilemma
- Students have opportunities to practice thoughtfully articulating their convictions

Ethical issue to be considered: data visualization, accessibility

Flow

- Students are reminded of what “ethics” is
- Students are introduced to the three ethical frameworks to be applied in future labs
- Students are asked to reflect on someone who they believe is of high moral character
- Students are asked how they believe their moral hero would respond to a situation based on their virtues
- Students consider the application of another ethical framework to the same situation

Preparation required:

- Read the entire lab several times to become familiar with the material.
- Print or post the student homework assignment.
- Consider questions that might be asked in class after the students have completed the assignment.

Here are two possibilities:

- Describe your moral hero and their most significant virtue
- To which scenario did you respond, and which framework was the most helpful?
Summarize your conclusion to the ethical dilemma based on that framework.

Bibliography